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DATA DESCRIPTOR

City-scale high-resolution traffic datasets with refined networks for hierarchical traffic control

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Urban-scale traffic control faces significant challenges due to the limited availability of large-scale and high-resolution datasets to support traffic control strategies across various network topologies and scales. To address this limitation, we present comprehensive traffic datasets for benchmarking traffic control strategies, possibly with hierarchical architectures. The datasets include five cities of varying scales with diverse intersection layouts and network topologies. We highlight the Xuancheng dataset, which adopts a refined network representation with hundreds of intersections, and achieves high spatiotemporal resolution by recording complete vehicle paths across the network. The dataset incorporates vehicle type information and provides historical trip data over long-term periods, with demand patterns exhibiting both periodic fluctuations and holiday-induced variations. Together with the refined road network and the high-resolution trip data, the dataset supports the construction of realistic traffic simulation environments. Furthermore, we derive multi-level traffic state data from the processed datasets, which facilitate the traffic control strategies spanning vehicular, lane-intersection, and network levels.

Background & Summary

Traffic congestion remains a persistent challenge in metropolises worldwide, leading to economic inefficiencies and environmental burdens¹. Addressing this challenge requires control strategies that coordinate decisions across multiple spatial and temporal scales, as both microscopic vehicle interactions and macroscopic network-level flow evolution shape traffic dynamics. In real-world traffic control practice, hierarchical traffic control architectures of different scales are often implemented as a prominent framework to alleviate traffic congestion². It integrates control strategies across multiple levels, including microscopic vehicular-level control, mesoscopic lane-intersection-level signal control, and macroscopic network-level perimeter control. Such architectures typically rely on heterogeneous traffic state data spanning multiple spatial scales^{3,4}. For example, Su *et al.*² integrated perimeter control based on network-level traffic states with local signal control driven by lane pressure⁵ to improve network throughput. Guo *et al.*³ leveraged vehicular-level state information (e.g., speed and acceleration) together with lane-level density to coordinate connected autonomous vehicles and signal control, resulting in reduced travel time and fuel consumption. However, high-quality traffic data capturing large-scale urban mobility, link-intersection traffic dynamics, and individual route-choice behaviors are still missing to empower hierarchical urban traffic control^{6,7}.

Existing open-source datasets for traffic control have several limitations. Firstly, most available datasets only cover networks with dozens of intersections in partial city areas, typically featuring simplified network topologies with predominantly four-approach intersections in regular grid structures^{8,9}. These datasets cannot capture the structural variety in real urban traffic networks, such as diverse intersection types, varying road segment lengths, and flexible junction topologies. Therefore, their capability to evaluate model generalizability across various urban network configurations is limited^{10,11}. Secondly, while some datasets encompass large networks with thousands of intersections, they often suffer from limited data collection periods, e.g., one hour^{3,12,13}. This limited temporal coverage restricts the diversity of traffic scenarios captured, particularly variations across

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weekdays and weekends, as well as peak and off-peak hours. As a result, control strategies developed on these temporally limited datasets may struggle to generalize to previously unseen traffic scenarios, leading to sub-optimal or unreliable performance when deployed in real-world systems with significant temporal variations. Thirdly, the sparse sampling rate of heterogeneous sensors restricts the spatio-temporal data resolution, thus confining the granularity of traffic flow analysis^{12,14}. Although some methods attempt to synthesize traffic trip data to cover the entire road network over extended periods¹³, the resulting route assignments fail to reflect the actual travel behavior and the travel preferences of drivers^{15,16}. Nevertheless, existing datasets exhibit limited representation of traffic heterogeneity by enforcing the homogeneous traffic assumption, e.g., uniform vehicle type, homogeneous traffic network configuration, and evenly distributed travel demand. These simplifications result in a notable mismatch between simulated environments and actual traffic conditions, eventually compromising the deployment of traffic control schemes from theory to practice. Given the limitations in network scale and topology, temporal coverage, spatio-temporal resolution, and traffic heterogeneity, existing datasets pose obstacles to developing robust and generalizable traffic control strategies for large-scale urban transportation systems.

To address the aforementioned challenges, we present city-scale high-resolution data to support hierarchical traffic control. The datasets are compiled using a data processing workflow that includes network reconstruction, signal standardization, and flow construction, transforming heterogeneous traffic records into a unified format. Using this workflow, we construct comprehensive datasets that span five cities of varying scales: Jinan, Hangzhou, Manhattan, Nanchang, and Xuancheng. Among them, the Xuancheng dataset is particularly notable for its refinement and resolution, encompassing hundreds of fine-grained intersections and providing highly sampled traffic trip data. The trip data is collected by automatic vehicle identification (AVI) over an extended period, where the AVI system classifies vehicles into different types. Building on this, we derive traffic state data from the datasets to support coordinated control actions across vehicular, lane-intersection, and network levels. Validation results demonstrate the effectiveness of the processing workflow in reconstructing network and trip data, and confirm that these city-scale high-resolution datasets enable the development and evaluation of hierarchical traffic control schemes.

Methods

In this paper, we offer five distinct real-world datasets spanning multiple scales and exhibiting various intersection layouts and network topologies. Each dataset integrates detailed road network topology, standardized traffic signal configurations, and calibrated traffic trip data, all constructed through a unified data processing workflow. These datasets facilitate the development of hierarchical traffic control approaches in urban networks. An overview of the data characterization, processing workflow, and data utilities is illustrated in Fig. 1.

Data acquisition and characterization. We adopt five datasets spanning diverse network scales and demand patterns to construct a benchmark source for hierarchical traffic control. As summarized in Table 1, these datasets vary in original data format, network refinement level, traffic volume, and data source, providing complementary coverage of both network topologies, and demand characteristics.

Data acquisition. The five datasets are collected through two distinct acquisition approaches. The Jinan, Hangzhou, Manhattan, and Nanchang datasets are obtained from publicly available sources^{8,13}. These datasets are primarily derived from the observations of traffic cameras, with both road network representations and trip-level traffic demand released in the CityFlow¹² and CBDData¹³ formats.

The Xuancheng dataset is constructed from a city-scale AVI system, with AVI sensors deployed at key intersections to capture vehicle passages across the network. As illustrated in Fig. 2, AVI observations cover most main roads in the study area. Among the 396 road segments without AVI observations, 394 correspond to one-lane branch roads with low traffic demand, while only two are main roads. This AVI system offers several advantages over camera-based observations. First, it records the entry and exit of nearly every vehicle within each road segment and matches vehicle identifiers across the network. Specifically, the entry table T_{entry} contains the fields (vehicle ID, vehicle type, road ID, entry time), while the exit table T_{exit} contains the fields (vehicle ID, vehicle type, road ID, exit time). By associating records from these two tables using anonymized vehicle identifiers, complete vehicle trips across multiple road segments can be reconstructed. Furthermore, the dataset enables concise classification of vehicles, providing explicit information on heterogeneous vehicle types. In addition to trip-level records, the road network of the Xuancheng dataset is derived from OpenStreetMap and pre-processed using official SUMO¹⁷ tools into the SUMO format.

Data characterization. As illustrated in Table 1, the datasets span multiple network scales, with the Xuancheng dataset providing a fine-grained city-scale representation. At the local scale, the Jinan dataset with 12 intersections and the Hangzhou dataset with 16 intersections represent typical arterial corridors. Due to their limited scales, these networks are widely utilized as benchmarks for vehicular-level control and signal control. At a larger spatial scale, the Manhattan dataset contains 48 intersections and exemplifies a grid-based urban network. To further examine scalability under increased network complexity, the Nanchang dataset expands the network size to 2,048 intersections, representing a city-scale scenario constructed using a coarse network representation. This dataset is suitable for examining the effectiveness of perimeter control in mitigating network-wide congestion. Distinct from these coarse datasets, the Xuancheng dataset comprises 791 intersections and adopts a refined network representation that captures detailed lane configurations, channelization, and complex intersection layouts. As shown in Fig. 3, this granularity enables a more realistic characterization of traffic dynamics and control interactions.

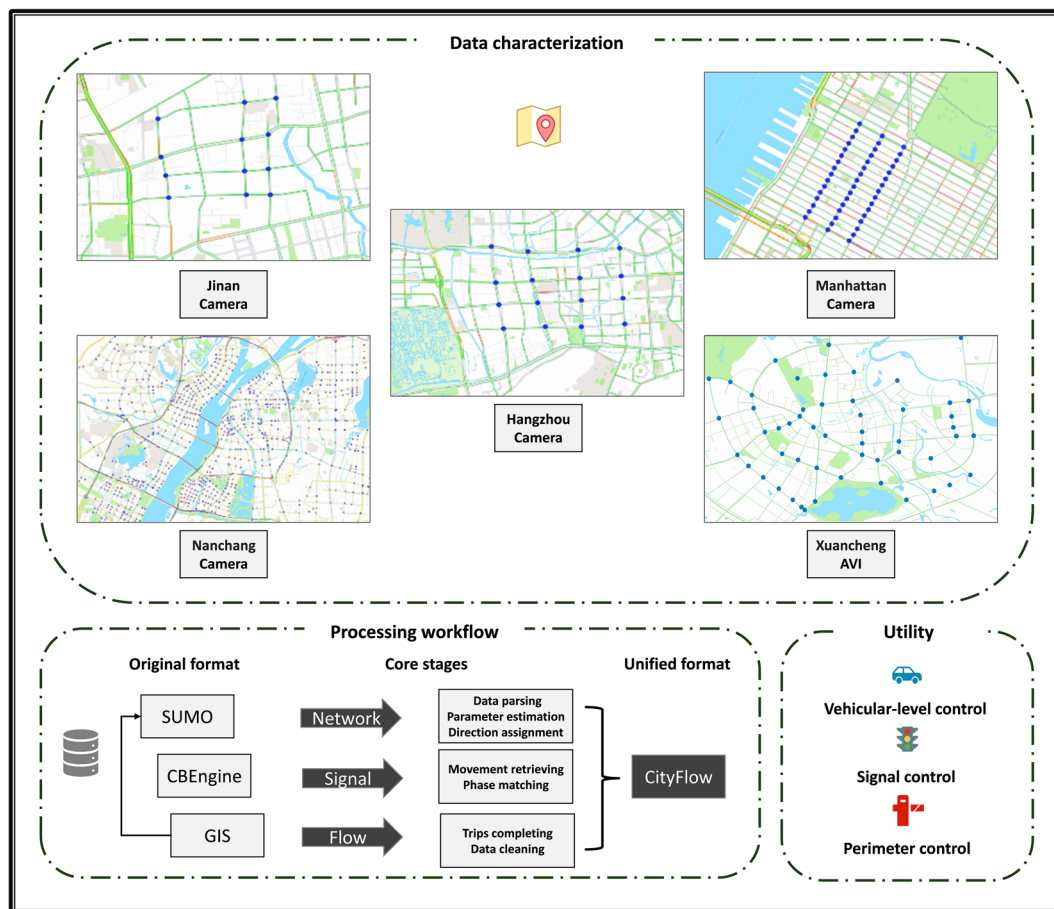


Fig. 1 Overview of the data characterization, processing workflow, and utilities of the datasets.

Attributes	Network Attributes			Demand Attributes		
City	Number of Intersections	Original Format	Network Refinement	Traffic Volume (trips)	Demand Duration	Data Source
Jinan ⁸	12	CityFlow	Coarse	6,295	1 hour	Camera
Hangzhou ²²	16	CityFlow	Coarse	2,983	1 hour	Camera
Manhattan ⁸	48	CityFlow	Coarse	2,824	1 hour	Camera
Nanchang ¹³	2,048	CBData	Coarse	9,876	1 hour	Camera
Xuancheng [*]	791	SUMO	Fine-grained	~ 350,000/day	1 month	AVI

Table 1. Illustration of the five traffic datasets, including network and demand attributes. ^{*}This paper is the first to present the Xuancheng dataset for hierarchical traffic control.

Beyond structural diversity, the datasets vary significantly in demand patterns and data sources. This variety enables analysis of both peak-hour dynamics and sustained demand fluctuations. Specifically, the Jinan, Hangzhou, Manhattan, and Nanchang datasets provide one-hour empirical intervals from traffic cameras, facilitating short-term analysis. In contrast, the Xuancheng dataset provides extended, month-long observations based on AVI records. Spanning one month with approximately 350,000 daily trips, it characterizes temporal variability across daily and weekly cycles, facilitating the development of hierarchical control under evolving conditions.

Unified workflow for city-scale data processing. The datasets are designed to support the simulation of hierarchical traffic control in large-scale urban networks, allowing the exploration of control strategies under diverse scenarios and reducing potential safety risks prior to real-world deployment. However, the incompatibilities among the diverse formats limit the direct applicability of the datasets in Table 1 and other similar datasets. To support reproducible and efficient traffic simulations, we adopt the open-source CityFlow simulator¹² as the core simulation environment and convert the source data into the CityFlow format. Among various well-established transportation simulation engines, such as Vissim¹⁸ and SUMO¹⁷, CityFlow is particularly well-suited for large-scale traffic simulation due to the following advantages:



Fig. 2 Observation coverage of the city-scale AVI system in the Xuancheng dataset.

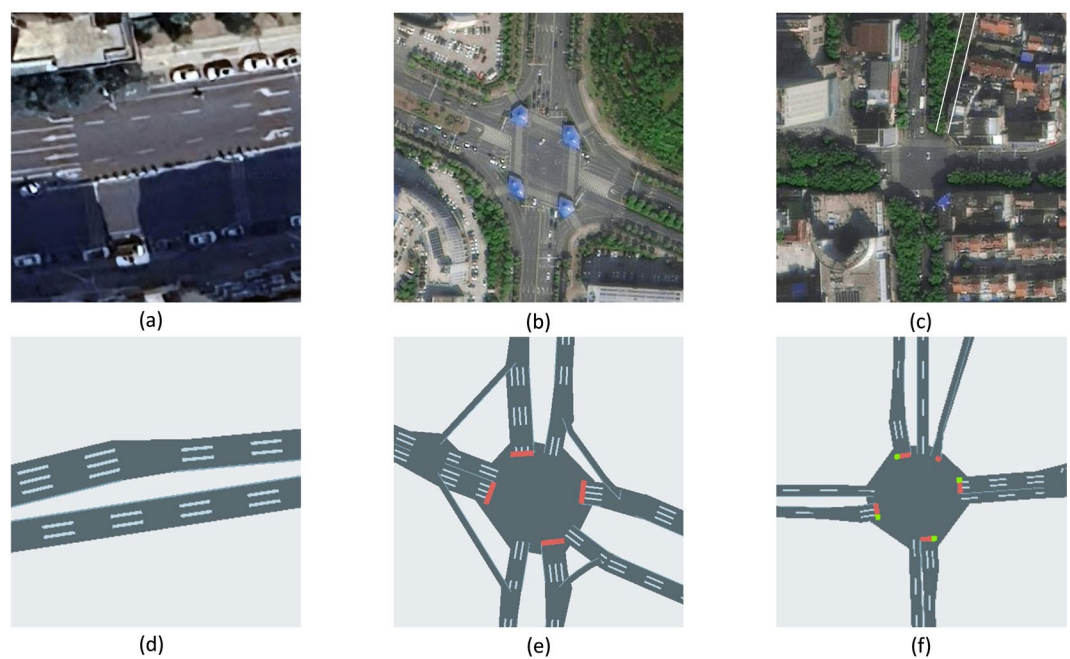


Fig. 3 Fine-grained network representations in the Xuancheng dataset. (a) Lane widening configurations, (b) channelization, and (c) complex intersection layouts (e.g. five-approach intersection) in the real-world network. (d)-(f) Corresponding network representations used in the dataset.

Element	Attribute	Format			
		GIS	CBData	SUMO	CityFlow
Intersection	Geographic coordinates	✓	✓	✓	✓
	Unique ID	✓	✓	✓	✓
	Signal control status	✓	✓	✓	✓
	Geometry (shape)	×	×	✓	✓
	Connected roads	×	×	✓	✓
Roads	Origin intersection ID	✓	✓	✓	✓
	Destination intersection ID	✓	✓	✓	✓
	Speed limit	✓	✓	✓	✓
	Lane configuration	✓	✓	✓	✓
	Road length	✓	✓	✓	✓
Traffic Signal	Controlled intersection ID	×	✓	✓	✓
	Incoming road mapping	×	✓	✓	✓
	Signal phase configuration	×	×	✓	✓
Movement	From-lane to to-lane	×	×	✓	✓
	Turning direction	×	×	✓	✓
	Movement index	×	×	✓	✓

Table 2. Attributes availability for traffic network elements across data formats.

- **Scalability** - It can reliably handle extensive urban networks with thousands of intersections while supporting high-volume traffic flows;
- **High performance** - It provides an interactive simulation environment with high computational efficiency;
- **Generalizability** - It includes user-friendly built-in functions along with generalizability for customized implementations. This is important for researchers to easily build system components for implementing hierarchical traffic control strategies.

To standardize heterogeneous traffic datasets into the CityFlow format, this paper proposes a unified workflow consisting of three core stages: road network reconstruction, traffic signal standardization, and flow construction.

Network reconstruction. Converting traffic network data into CityFlow format presents two major challenges: (1) the diversity of source formats and (2) accurate reconstruction of refined road networks. Firstly, to accommodate diverse input formats, including SUMO, CBData, and Geographic Information Systems (GIS), we develop a data parsing component capable of extracting structural elements and inferring topological relationships across formats. This component identifies intersections (nodes) and roads (edges), and accurately maps their connectivity, thereby establishing a consistent foundation for subsequent network reconstruction. Secondly, to reconstruct refined road networks that contain detailed lane configurations, channelization, and diverse intersection layouts, we incorporate parameter estimation and direction assignment to model complex intersections. These components handle multi-approach intersections with diverse geometric layouts, as well as bidirectional intersections with varying lane numbers and segment channelization.

The first component of the reconstruction workflow is data parsing, which aims to extract the core structural elements of road networks, specifically intersections (nodes) and roads (edges), from heterogeneous source formats. The elements available in each format are summarized in Table 2. Among these formats, GIS data is primarily used as a base map, providing spatial and semantic information. However, it lacks explicit representations of dynamic traffic elements such as movements and signal phase logic. In contrast, CBData and SUMO are designed for traffic simulation and encode additional elements such as movements and signal configurations. However, CBData does not provide geometric representations of intersections, and its signal control is limited to directional permissions without defining turning movements. SUMO offers a more detailed and modular representation of intersections, roads, movements, and traffic signals, yet still requires inferring topological relationships to construct a coherent network graph. The core task in this component is to construct the network's topological structure. The traffic network is represented as a directed graph $G = (V, E)$, where V denotes the set of intersections (nodes), and E denotes the set of directed roads (edges) that capture connectivity among the intersections. For any intersection $v \in V$, its incoming and outgoing roads are defined as:

$$\text{In}(v) = \{e \in E | e = (u, v, a_e)\}, \quad \text{Out}(v) = \{e \in E | e = (v, w, a_e)\} \quad (1)$$

where each road $e \in E$ is represented as a triplet $(v^{\text{start}}, v^{\text{end}}, a_e)$, in which v^{start} and v^{end} denote the origin and destination intersections, and a_e is the set of edge attributes, such as speed limit, number of lanes, segment length, and geometric shape. Based on this formulation, a complete graph representation of the road network in CityFlow can be constructed by parsing intersections and roads from each input format and aligning their topological relationships.

The second component of the reconstruction workflow is parameter estimation, which focuses on reconstructing intersection geometries and estimating their associated parameters. Two limitations hinder the direct

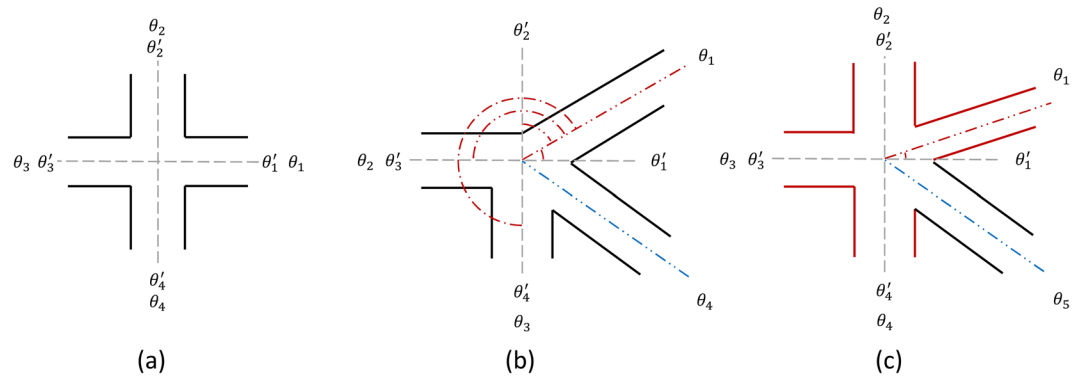


Fig. 4 Direction assignment for various intersection layouts. **(a)** Standard intersection. **(b)** The absolute angular deviation of approach θ_1 . **(c)** Direction assignment for a five-approach intersection.

use of topology parameters derived from raw geometric boundaries. First, traffic simulators employ different intersection modeling approaches, resulting in inconsistent definitions of geometric attributes. For example, CityFlow characterizes an intersection using a center point and a width, while SUMO represents an intersection using boundary points extracted from stop-line geometries. Second, existing conversion tools often rely on simplified assumptions that fail to capture the variability and geometric complexity of real-world intersections. For instance, the official SUMO-to-CityFlow converter assigns a uniform width of 15 meters to all intersections. To address these issues, we first apply convex hulls to normalize heterogeneous intersection geometries, converting irregular boundaries into closed polygons. Then, minimum bounding rectangles are derived from the hulls to estimate the parameters. For each intersection, boundary points $\mathbf{Q} = \{p_1, p_2, \dots, p_n\}$ are extracted from the source data, and a convex hull is computed. Formally, the convex hull is defined as:

$$\text{Conv}(\mathbf{Q}) = \left\{ \sum_{i=1}^n \lambda_i p_i \mid \lambda_i \geq 0, \sum_{i=1}^n \lambda_i = 1 \right\} \quad (2)$$

where λ_i represents the weight assigned to each point p_i in the convex combination, and $\lambda_i \geq 0$ ensures non-negativity of the weights. The constraint $\sum_{i=1}^n \lambda_i = 1$ guarantees that the resulting point lies within the convex hull defined by the set of points $\mathbf{Q}$. To estimate the parameters of each intersection, we compute the minimum bounding rectangles of the convex hull using *shapely.geometry.Polygon*. The effective width and center point of the intersection are then derived from this rectangle.

The third component of the reconstruction workflow is direction assignment, which is essential for movement classification and consistent phase design. Existing methods typically assign directions based on interval, angle ordering, and angle difference¹⁹. However, these rule-based methods often perform poorly when applied to real intersections and fail to handle cases with three or five approaches. We formulate direction assignment as a global optimization problem, which minimizes the total absolute angular deviation between actual approaches and their assigned cardinal directions. A standard intersection is characterized by four approaches aligned with the cardinal directions, as illustrated in Fig. 4a. For a real-world four-approach intersection (e.g., Fig. 3b), we compute the absolute angular deviation between each approach and each cardinal direction (see Fig. 4b), and determine the assignment that minimizes the total deviation. This formulation can be easily extended to intersections with fewer approaches, since cardinal directions could remain unassigned without altering the optimization structure. For five-approach intersections (e.g., Fig. 3c), we select the four approaches that yield the minimum total angular deviation (as illustrated by the red approaches in Fig. 4c). The assignment problem is formulated as:

$$\min \sum_{i=1}^k \sum_{j=1}^4 |\theta_i - \theta'_j| x_{ij} \quad (3)$$

subject to

$$\sum_{i=1}^k x_{ij} \leq 1, \quad j = 1, \dots, 4 \quad (4a)$$

$$\sum_{j=1}^4 x_{ij} = 1, \quad i = 1, \dots, k \quad (4b)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \quad (4c)$$

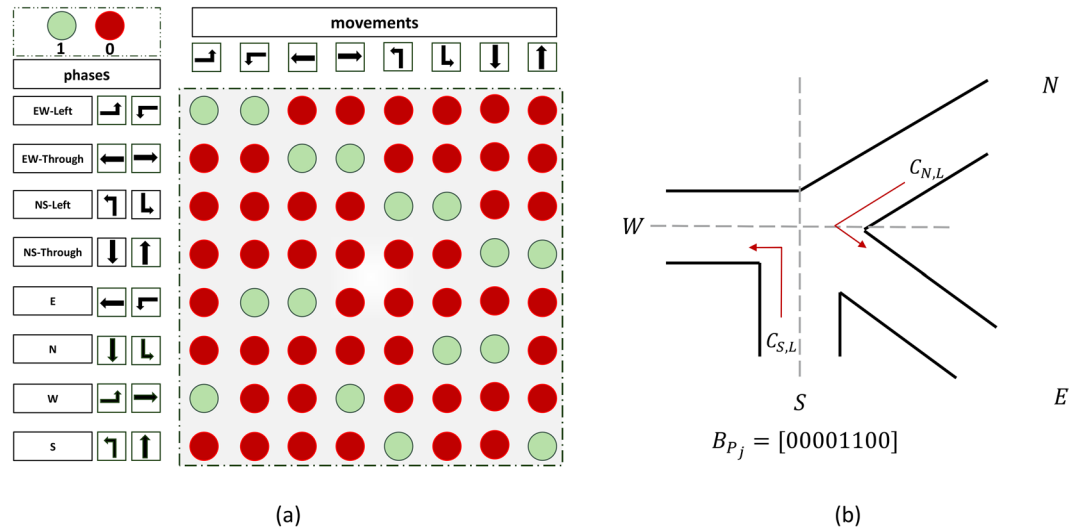


Fig. 5 Phase table linking signal phases to permitted movements and an example phase design case. **(a)** Phase table. **(b)** Case of the standardized phase design.

where θ_i is the angle of the i -th approach, θ'_j is the standard angle of the j -th cardinal direction, x_{ij} is the decision variable. In particular, k is the number of approaches while 4 is the number of cardinal directions, and the two are not always equal. Equation (4a) specifies that each direction can be assigned to at most one approach, while Equation (4b) ensures that each approach is assigned exactly one direction. Equation (4c) ensures the binary decision variable, where $x_{ij} = 1$ indicates that direction j is assigned to approach i . To solve this optimization problem, we transform it into a cost matrix and employ the Hungarian algorithm²⁰.

By integrating data parsing, parameter estimation, and direction assignment, the network reconstruction achieves robust and accurate conversion of heterogeneous traffic datasets. This modular architecture enables consistent representation of complex intersection geometries and lane configurations, supporting scalable city-scale simulation.

Signal standardization. Signal phase data serve as the basis for both intersection coordination and higher-level strategies, such as perimeter control²¹. However, designing phases for hundreds of intersections in a city-scale network requires significant effort. We propose a standardized phase design that unifies movement classification and automates phase matching. It can significantly reduce manual configuration across diverse intersection layouts. The design consists of two steps: (1) retrieving the sequence index of available movements and (2) selecting feasible signal phases based on the layout of each intersection.

To retrieve the sequence index of available movements, we apply movement classification at each intersection to derive all permissible movements based on the direction assignment described above. Let $D = \{E, S, W, N\}$ denote the set of directions, and let $M = \{L, T\}$ represent left-turn and through movements (right-turn movements are treated as free-flow and excluded from signal phase data). The set of controlled movements is defined as:

$$C = \{c_{d,m} | d \in D, m \in M\}, \quad |C| = 8 \quad (5)$$

where $c_{d,m}$ represents the controlled movement from direction d with movement type m . Based on the controlled movements, we can identify the corresponding movement indices.

Based on the eight controlled movements defined above, we construct eight standardized phases that cover the majority of real-world signal control scenarios, as illustrated in Fig. 5a. These phases are organized into movement-based groups: the first four represent the left-turn and through movements for the east-west and north-south directions, and the last four correspond to individual directional movements. The eight phases are as follows: EW-Left, EW-Through, NS-Left, NS-Through, E, N, W, S, denoted as:

$$\mathcal{P} = \{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8\} \quad (6)$$

where $P_1 = \{c_{E,L}, c_{W,L}\}$ represents the left-turn movements along the east-west direction, $P_2 = \{c_{E,T}, c_{W,T}\}$ represents the through movements along the east-west direction, and so on.

To ensure the phase design accommodates variations in intersection layouts, we develop a phase matching mechanism to select appropriate standardized phases for existing configurations. Each standardized phase P_i is represented as a binary vector B_{P_i} , where 1 indicates an allowed movement, and 0 indicates a forbidden movement. For a target phase P_j in the intersection configurations, we compute the L_2 distance between its phase vector and each standardized phase vector and select the phase with the minimal value:

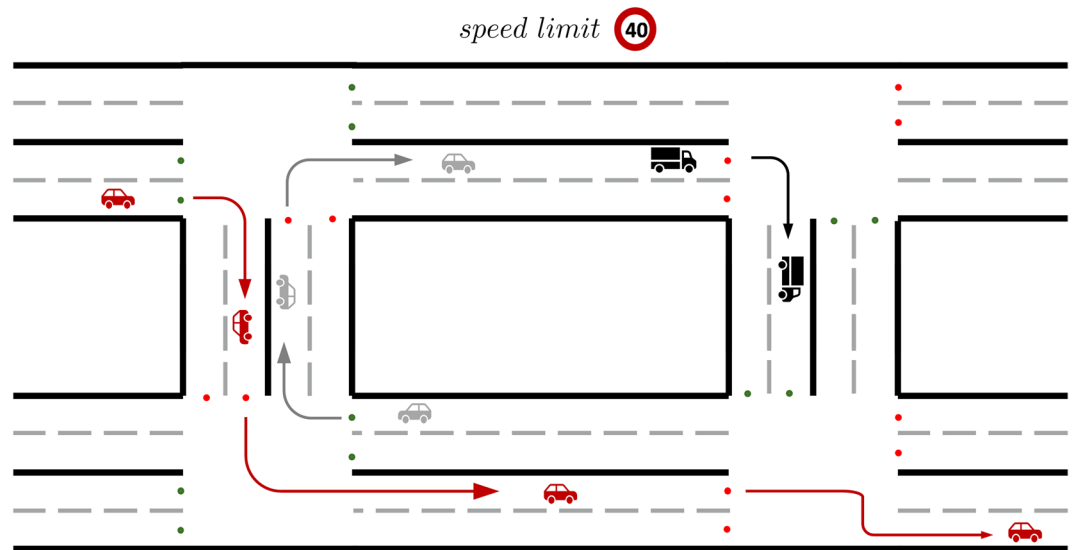


Fig. 6 Illustration of integrating multiple records to reconstruct complete trip paths for individual vehicles.

$$i^* = \arg \min_i \|B_{P_i} - B_{P_j}\|_2 \quad (7)$$

This approach preserves the original control logic across diverse intersection layouts while enabling adaptation to the standardized phases, ensuring compatibility with both manually designed and automatically generated phase plans. As an illustrative example, consider the four-approach intersection in Fig. 5b, where the controlled movements include $c_{S,L}$ and $c_{N,L}$, representing the southbound and northbound left-turn movements, respectively. During phase matching, the binary encoding of the observed intersection phase P_j yields the minimum L_2 distance from B_{P_3} . Consequently, the third standardized phase P_3 , shown in Fig. 5a, is identified as the matched phase.

Flow construction. Existing public traffic datasets often suffer from limited temporal duration, which restrains both the analysis of urban traffic patterns and the training of data-driven algorithms. Datasets such as Jinan¹², Hangzhou²², and Manhattan¹² cover only partial metropolitan networks and span just one hour of observations derived from camera data. With such short durations, traffic phenomena such as congestion propagation and spillover effects cannot be adequately captured. Therefore, city-scale datasets with sufficient observation periods are essential to capture the complex dynamics of urban traffic systems²³. Although the Nanchang dataset¹³ contains a road network with more than 2,000 intersections, the sampled traffic volume within one hour includes only 9,786 vehicles, which may be insufficient to reflect the spatial and temporal variability of traffic demand across a large network.

To remedy this, the Xuancheng dataset was collected over one month, enabling the analysis of day-to-day and weekly traffic patterns. By equipping AVI sensors at the entrances and exits of nearly all road segments in Xuancheng^{24,25}, the city generates high-resolution traffic data amounting to approximately 4,000,000 records per day. In contrast to camera-based approaches that rely on predefined turning ratios to estimate complete trip paths²², the AVI-based approach directly records the movements of almost every vehicle within each road segment²⁶. Therefore, by tracking each vehicle across multiple road segments, as illustrated in Fig. 6, we aggregate millions of flow records to derive around 350,000 trip data points per day. We first join the vehicle entry table T_{entry} and exit table T_{exit} to construct road-level passage records:

$$L = T_{\text{entry}} \bowtie \begin{matrix} T_{\text{entry}}.\text{vehicle ID} = T_{\text{exit}}.\text{vehicle ID} \\ T_{\text{entry}}.\text{road ID} = T_{\text{exit}}.\text{road ID} \\ 0 < T_{\text{exit}}.\text{exit time} - T_{\text{entry}}.\text{entry time} \leq 300s \end{matrix} T_{\text{exit}} \quad (8)$$

where the entry table T_{entry} contains the fields (vehicle ID, vehicle type, road ID, entry time) and the exit table T_{exit} contains the fields (vehicle ID, vehicle type, road ID, exit time). The resulting set L consists of integrated passage records with the schema (vehicle ID, vehicle type, road ID, entry time, exit time). The join operation $\bowtie$ matches entry and exit records for the same vehicle within the same road, subject to a logical temporal order and a maximum travel time of 300 seconds.

Next, we reconstruct complete vehicle trips by aggregating consecutive passage records from L , ensuring that the time gap between adjacent records does not exceed 30 minutes:

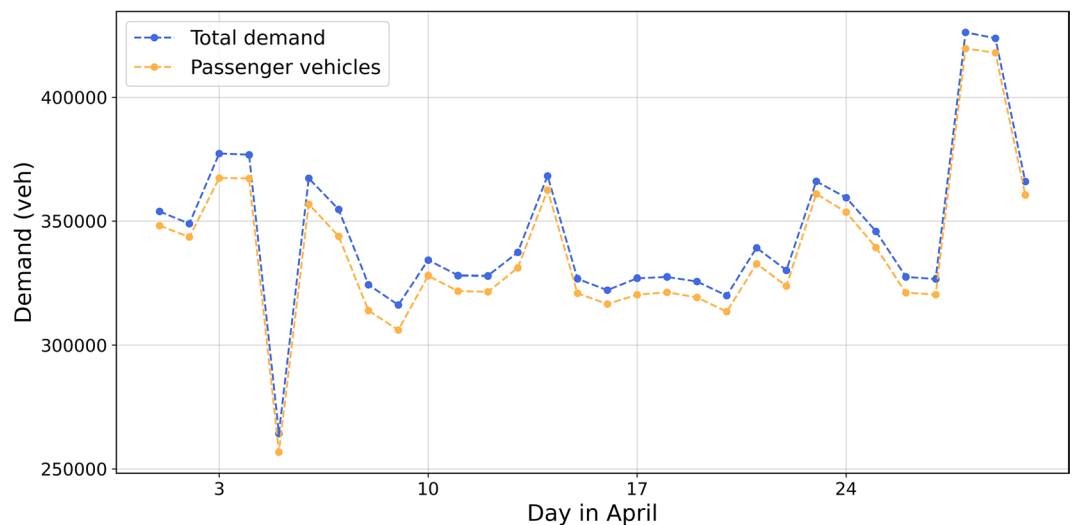


Fig. 7 Daily aggregated trip demand in Xuancheng over April 2023.

$$T = \{(n, l_1, l_2, \dots, l_r) | l_i \in L, l_i \text{ vehicle ID} = n, \\ \forall i \in [1, r-1]: l_{i+1} \text{ exit time} - l_i \text{ exit time} \leq 1800s\} \quad (9)$$

where $l_i \in L$ represents a valid passage record. The resulting set T consists of reconstructed vehicle trips, represented as ordered sequences of segment passages associated with a unique vehicle ID n , thus capturing continuous travel behavior across the network. Notably, all privacy-sensitive information, such as license plate numbers, has been anonymized in the trip data.

Furthermore, we incorporate vehicle type information into the trip data, which classifies vehicles into two categories: passenger vehicles and freight vehicles. Specifically, passenger vehicles constitute 94% of the total traffic volume, as shown in Fig. 7, and are primarily standard passenger vehicles with a small proportion of large buses. Freight vehicles account for the remaining 6%, mainly consisting of light trucks, while heavy trucks are rarely observed due to access restrictions. By incorporating vehicle type information, the traffic simulation more accurately reflects real-world vehicle composition. This classification also enables a more realistic representation of how buses and trucks influence traffic flow dynamics, particularly given their distinct acceleration and braking behavior.

To ensure the reliability of constructed trips, a data cleaning procedure is typically applied before finalizing the trip dataset²⁷. First, trips involving maneuvers inconsistent with the road network, such as turning movements prohibited at specific intersections, are identified as invalid and excluded. Second, trips with missing or incomplete AVI passage records are repaired via route inference. This process leverages CityFlow's Dijkstra-based routing module to compute the shortest feasible path consistent with the predefined road topology.

Figure 7 presents the daily aggregated trip demand in Xuancheng over April 2023. The demand profile exhibits periodic fluctuations as well as holiday-induced variations. On typical days, daily demand ranges from 320,000 to 370,000 *veh*, with pronounced deviations during statutory holidays (e.g., April 5 and April 28–29). The tick marks at days 3, 10, 17, and 24 denote Mondays in April, suggesting a recurring weekly demand pattern. In particular, demand drops to around 260,000 *veh* on April 5 and increases to approximately 420,000 *veh* during April 28–29.

Utility of multi-scale traffic data. The reconstructed datasets provide the key information needed for hierarchical traffic control, including network topology and trip data. The collection of datasets differs in their network and demand characteristics, offering complementary strengths in spatial resolution, temporal coverage, and network scale. For example, the Jinan and Hangzhou datasets focus on small-scale arterial corridors, while the Manhattan dataset represents a medium-scale grid network, and the Nanchang dataset captures city-scale traffic dynamics using a coarse network representation. In contrast, the Xuancheng dataset provides fine-grained network topology and long-term reconstructed vehicle trips derived from AVI records. Collectively, they span a wide range of urban traffic conditions, from small grid networks to city-scale networks, and from short-term peak observations to long-term demand dynamics. This diversity enables the construction of realistic simulation environments for the development and validation of advanced traffic control strategies. We replay the trip data in a traffic simulation environment to derive traffic state data that characterizes traffic dynamics, serving as the basis for design and evaluation of hierarchical control²¹.

Traffic state data. Traffic state data, derived from the trip data, enable real-time traffic monitoring and state estimation²⁸. These data are organized into three interconnected levels: vehicular-level, lane-level, and

Attribute				
Level	Speed	Flow	Density	Others
Vehicular-level	✓	×	×	Position, trip paths, leading vehicle
Lane-level	✓	✓	✓	Lane length, speed limit, queue length
Network-level	✓	✓	✓	Road network topology, vehicle ID set

Table 3. Attributes captured at different state data levels.

network-level, capturing traffic dynamics from microscopic to macroscopic scales. Vehicular-level state data are aggregated to form lane-level traffic states, which are further aggregated into network-level states. The attributes available at each level are summarized in Table 3.

- Vehicular-level state data provide microscopic attributes such as speeds, trip paths, and positions of individual vehicles, which are essential for routing control. Incorporating leading vehicle information further enables advanced strategies such as platooning^{29,30}. These attributes serve both as inputs to vehicular-level control and as the basis for higher-level aggregation. Furthermore, vehicular-level state data further supports the evaluation of trajectories generated by subsequent control strategies.
- Lane-level state data aggregates vehicular-level data to represent mesoscopic traffic states. It captures structural attributes such as lane length and speed limit, as well as dynamic attributes including queue length, average speed, flow, and density. These states serve as key inputs for signal control, supporting signal timing, phase optimization³¹.
- Network-level state data provides a macroscopic view of traffic by aggregating traffic states across a regional network. It includes both static attributes, such as network topology and connectivity, and dynamic attributes that reflect aggregate traffic conditions. Network-level state data can further support traffic bottleneck analysis and large-scale coordination via traffic control^{32–34} or vehicle dispatching³⁵.

Such a state representation naturally supports hierarchical traffic control strategies operating at vehicular, lane-intersection, and network levels. Vehicular-level control regulates individual vehicle maneuvers, lane-intersection-level control governs signal timing and phase operations^{21,36,37}, and network-level control regulates traffic inflows into protected regions to prevent oversaturation^{38–40}.

Data Records

We share the traffic datasets on a public repository (Figshare⁴¹). The datasets for each city are provided in two formats: (1) source data after anonymization, and (2) CityFlow-formatted data. To facilitate reproducibility and enable participants to readily utilize the data, the CityFlow-formatted data, released in JavaScript Object Notation (JSON), contains the road network and traffic trip data. The detailed attributes of the road network and traffic trip data are shown in Fig. 8. The road network comprises two core structures: intersections and roads, each describing the geometric and topological attributes of the urban network. Their corresponding attributes are listed in Table 4. The traffic trip data contains vehicle-related information, including vehicle characteristics, departure time, and route. Vehicle characteristics include static, dynamic, and safety parameters. The detailed attributes of the traffic trip data are summarized in Table 5.

To support customized preprocessing and enable flexible data use, source data after anonymization are also provided. Although the specific source data formats vary across cities, as shown in Table 1, they still include both road network and trip data, and generally follow a structure compatible with that used in CityFlow.

Technical Validation

This study provides city-scale, high-resolution data that support hierarchical traffic control strategies. To ensure consistency between the converted data and real-world traffic conditions, we validate the outputs from two perspectives: network data and trip data. Furthermore, we conduct a series of experiments to demonstrate the feasibility of using traffic state data. All experiments are carried out using the Xuancheng dataset, which provides a realistic city-scale environment for validation. The Xuancheng network comprises hundreds of signalized intersections and thousands of road segments, encompassing a diverse range of road types and intersection layouts. This dataset also provides city-scale traffic demand data, which allows the simulation to reproduce realistic traffic volumes and urban traffic dynamics.

The computational setup consists of a workstation equipped with an AMD Ryzen 7 5800X CPU and 64 GB of RAM. Simulating one hour of traffic signal control in the Jinan network at a one-second resolution requires roughly half a minute of computation. When scaling to larger scenarios with hierarchical traffic control, such as simulating signal control and perimeter control for one hour across the entire Xuancheng network at a one-second resolution, the computation requires approximately five minutes. Additionally, simulations involving multiple cities can be executed in parallel to distribute the computational workload.

Network data validation. While the proposed processing workflow enables accurate reconstruction of fine-grained road networks from diverse source formats, verifying the reliability of the outputs remains essential. To ensure data quality, we validate the Xuancheng network along two dimensions: network consistency and node consistency.

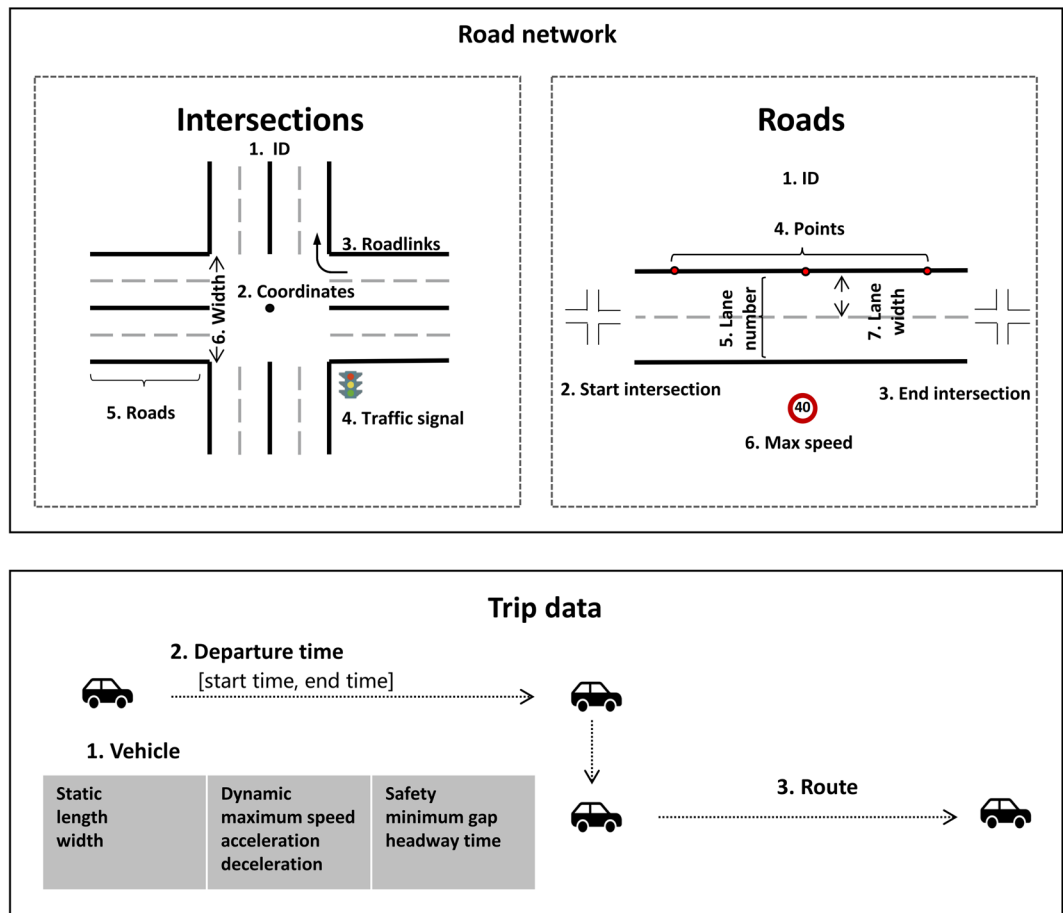


Fig. 8 Attribute descriptions of road network and trip data.

Section	Attribute	Description
Intersection	Intersection ID	Unique identifier of the intersection.
	Coordinates	Spatial coordinates of the intersection center (x, y).
	Road links	List of permitted movements at the intersection.
	Traffic signal	Traffic signal settings of the intersection.
	Connected roads	Incoming and outgoing roads associated with the intersection.
	Width	Physical width of the intersection (m).
Road	Road ID	Unique identifier of the road.
	Start intersection	ID of the upstream intersection.
	End intersection	ID of the downstream intersection.
	Geometry points	Sequence of points describing the road alignment.
	Lane attributes	Lane configuration, including lane width (m) and lane count.
	Maximum speed	Maximum allowable speed on the road (m/s).

Table 4. Description of the road network.

We first assess the network consistency between the converted network and the original GIS data using the Jaccard similarity, which quantifies the preservation of connectivity patterns and local structural relationships. The Jaccard similarity is calculated as follows:

$$J(v_i^O, v_i^C) = \frac{|\mathcal{N}(v_i^O) \cap \mathcal{N}(v_i^C)|}{|\mathcal{N}(v_i^O) \cup \mathcal{N}(v_i^C)|} \quad (10)$$

$$J = \frac{1}{|V|} \sum_{i=1}^{|V|} J(v_i^O, v_i^C), \quad (11)$$

Section	Attribute	Description
Vehicle	Vehicle length	Physical length of the vehicle (m).
	Vehicle width	Physical width of the vehicle (m).
	Maximum acceleration	Maximum achievable acceleration (m/s ²).
	Normal acceleration	Normal/desired acceleration during typical driving (m/s ²).
	Maximum deceleration	Maximum achievable deceleration (m/s ²).
	Normal deceleration	Normal/desired deceleration during typical driving (m/s ²).
	Maximum speed	Maximum permissible speed for the vehicle (m/s).
	Minimum gap	Minimum desired spacing to the front vehicle (m).
	Headway time	Duration to the front vehicle (s).
Trip	Route	Sequence of road IDs followed by the vehicle.
	Departure time	Vehicle generation time (s), defined within a periodic interval.

Table 5. Description of the traffic trip data.

where v_i^O and v_i^C represent the corresponding intersection in v the original and converted networks, and $\mathcal{N}(v_i^O)$ and $\mathcal{N}(v_i^C)$ represent their respective neighbor sets. J is the average Jaccard similarity. A higher similarity indicates better preservation of adjacency relationships and local intersection structure. The average Jaccard similarity is 0.87, indicating a high level of topological consistency between the converted and original networks, confirming the reliability of the conversion process. An illustrative comparison is shown in Fig. 9.

We then evaluate node consistency using both positional displacement and node parameters. The mean Euclidean distance (MED) is employed to quantify positional deviations introduced during network conversion. It is defined as the average Euclidean distance between corresponding nodes in the converted network and the original GIS network:

$$\text{MED} = \frac{1}{|V|} \sum_{i=1}^{|V|} \|\mathbf{p}_i - \mathbf{p}_i^{\text{source}}\|, \quad (12)$$

where $\mathbf{p}_i$ and $\mathbf{p}_i^{\text{source}}$ denote the coordinates of the i -th intersection v in the converted network and the GIS data, respectively. The resulting MED of 3.58 meters indicates that the average positional deviation remains within an acceptable range, confirming the spatial accuracy of our network conversion.

Building upon the consistency assessment, we further validate the reconstruction of intersection geometric parameters, as illustrated in Fig. 10. The dark red region represents the SUMO-based intersection, from which the boundary points (blue points) are extracted. We first compute the convex hull (green lines) of these boundary points to obtain a compact polygon. The convex hull primarily captures the stop-line geometry while excluding the channelized right-turn islands, yielding a shape that aligns well with the CityFlow representation. We then compute the minimum bounding rectangle (light red region) of the convex hull to derive estimates of the intersection center and width.

Trip data validation. The converted trip data are validated by comparing simulated travel times with their real-world counterparts. To mitigate the impact of external variability, a road segment of approximately 560 meters is selected, with traffic data recorded before 6:00 AM. The selected segment is a single-lane road that widens to two lanes only at the downstream intersection approach. The upstream and downstream intersections comprise three and four approaches, respectively, and both operate under a Webster-based signal control scheme. The single-lane configuration eliminates the influence of lane-changing maneuvers, thereby enabling a more accurate assessment of car-following behavior and travel time consistency between the simulation and observations.

We reconstruct the real-world traffic environment by replicating empirical signal control protocols and assigning trip paths consistent with observed real-world movements. As shown in Fig. 11, the simulated travel times closely match the real-world observations. Specifically, the average simulated travel time is 78.5 seconds, while the observed average travel time is 76.6 seconds. The discrepancy can be attributed to two primary factors. First, in real-world conditions, drivers may accelerate near the end of the green phase, a behavior not represented in the simulation. Second, the simulation enforces a strict speed limit constraint, which ensures that simulated minimum travel times reflect ideal free-flow conditions. In contrast, real-world drivers may exceed posted speed limits, producing shorter travel times than the simulated minimum.

State data validation. Experiments are conducted to demonstrate the feasibility of using traffic state data derived from the Xuancheng dataset for hierarchical traffic control.

We choose the max-pressure signal control strategy^{5,42}, a benchmark method for managing saturated urban networks, as an example control scenario using the lane-level state data. Each intersection deploys max-pressure control during the evening peak hour (5:00–6:00 PM). The impact of signal control on the spatial distribution of congestion is illustrated in Fig. 12a, where several intersections exhibit higher density levels (red areas). By analyzing network-level state data, perimeter control is introduced to regulate traffic at critical boundary intersections with high inflow demand. The spatial density distribution in Fig. 12b further demonstrates a shift toward lower density levels (green coloring), particularly within the area marked by the black dashed box.

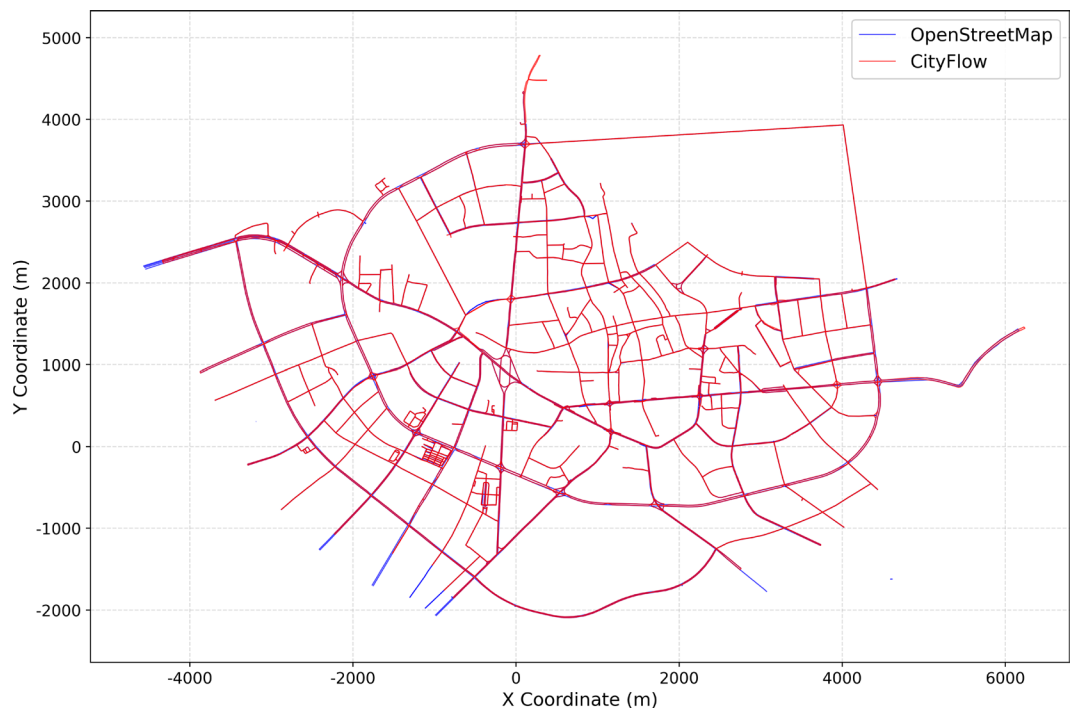


Fig. 9 Spatial deviations between converted and original GIS networks.

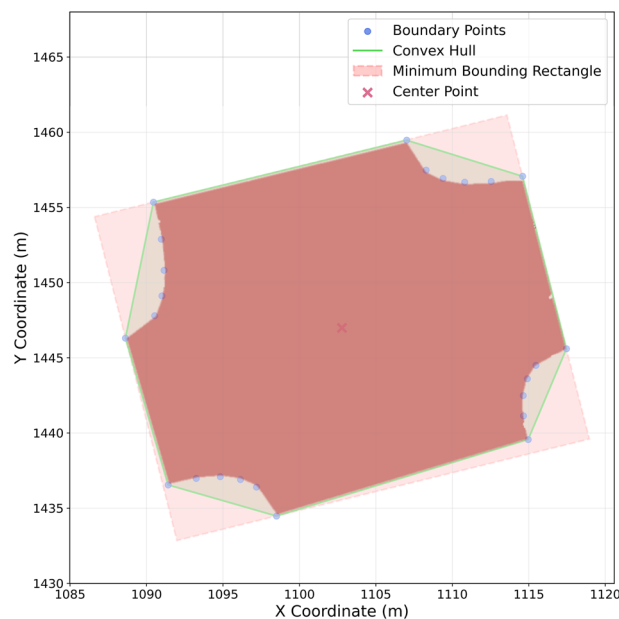


Fig. 10 Procedure of parameter estimation.

Furthermore, vehicular-level control is implemented to provide microscopic management within the hierarchical control. A case study considers a single vehicle traveling from a designated origin to a destination within the Xuancheng network along a high-volume corridor, represented by the black straight-through path in Fig. 12a. The vehicle following this path has not completed the trip after 400 s due to congestion along the corridor. By contrast, the alternative grey path in Fig. 12a, despite its longer distance, experiences lighter traffic conditions and reaches the destination at around 350 s. These results demonstrate that the traffic state data derived from the Xuancheng dataset can effectively support the development and evaluation of hierarchical traffic control strategies.

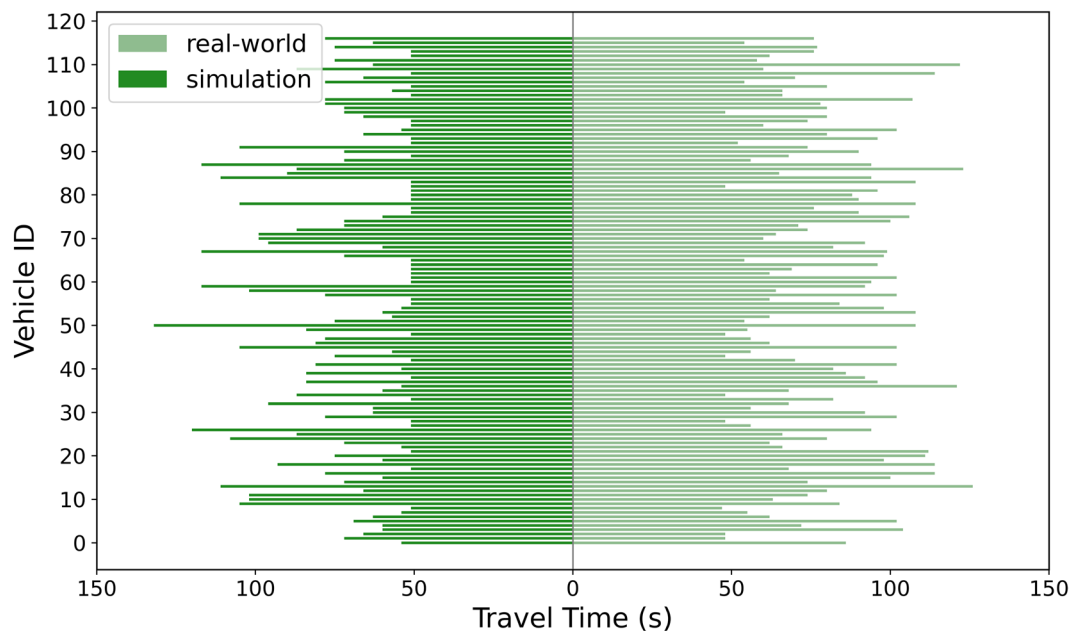


Fig. 11 Travel time from simulation and observation on the selected road segment.

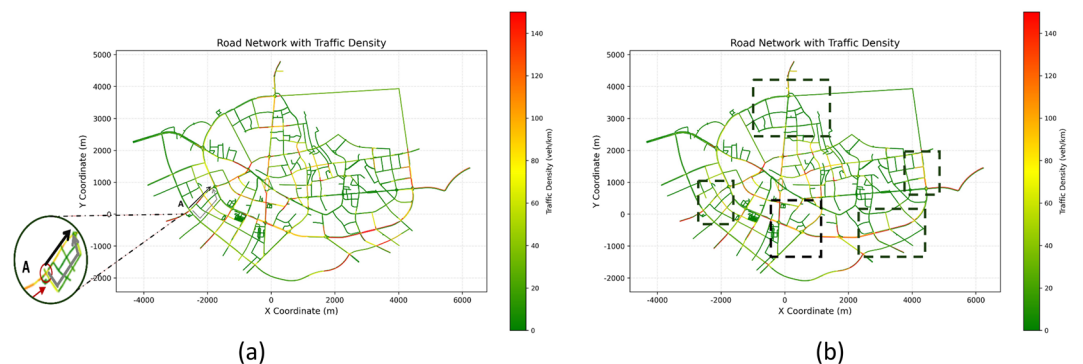


Fig. 12 Comparison of traffic density distributions with and without perimeter control. (a) Signal control. (b) Signal control with perimeter control.

Usage Notes

Each city dataset consists of three main files: a configuration file, a road network file, and a traffic trip file. The configuration file allows researchers to customize key simulation parameters, including the simulation time step, phase activation status, and specify the road network and trip files to be loaded. Details of the road network and traffic trip files are provided in the Data Records section. The datasets are organized into two main directories: *cfg* and *data*. The *cfg* directory contains multiple subfolders, each corresponding to a specific dataset and storing its associated configuration files for that dataset. The *data* directory contains subfolders for each dataset, in which the associated road network file and traffic trip file are stored together to define the static network topology and traffic demand.

To facilitate seamless adoption and reproducibility, we provide detailed online usage documentation in Figshare⁴¹. The documentation introduces a hierarchical traffic control architecture spanning vehicular, lane-intersection, and network levels, and describes the operational logic of each control level. It further presents the formulation and computation procedures of the proposed multidimensional performance metrics across different levels. In addition, representative case studies based on the Xuancheng dataset are provided to demonstrate the practical use of multidimensional evaluation in hierarchical traffic control. Specifically, vehicular-level control is evaluated using vehicle trajectories and travel time, lane-intersection-level control is assessed using density, outflow, and delay, and network-level control is evaluated through MFD-based metrics.

Data availability

The dataset is available at Figshare⁴¹.

Code availability

Our code used in this paper for the dataset setup, data analysis, and experiments can be found in GitHub (https://github.com/sysu-mqz/Hierarchical_traffic_control_platform).

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Author contributions

Qinzhou Ma, Weifan Zhong, and Xinling Guo conducted data collection and data processing; Qinzhou Ma, Renxin Zhong, and Wenfei Ma carried out technical validation; Qinzhou Ma, Renxin Zhong, and Wenfei Ma drafted the original manuscript; Renxin Zhong, Zicheng Su, Zhaocheng He, and Wenfei Ma contributed to the review and editing of the manuscript; supervision was provided by Renxin Zhong; project administration and funding acquisition were handled by Renxin Zhong. All authors have read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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